

Gaming Caps with Fixed-Cost Circumvention

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Abstract

How does tightening a quantity cap affect total consumption, and which users account for the resulting reductions, when unrestricted access can be restored at a fixed private cost? Motivated by incomplete compliance with online gaming restrictions, this paper studies heterogeneous demand and fixed-cost circumvention. Intermediate-demand users remain constrained, while higher-demand users pay to recover unrestricted consumption. Tightening reduces use among users who remain constrained but induces others to circumvent. Holding the cost and demand distribution fixed, a marginal tightening raises aggregate consumption to first order at an interior configuration exactly when density at the circumvention threshold exceeds its average over the constrained type interval. In the quadratic benchmark with uniform types, tightening within the interior region leaves aggregate consumption unchanged while shifting reductions toward lower-demand users. The policy-weighted reduction therefore weakly falls for every continuous, nonnegative, nondecreasing weight on demand. The density condition extends to preferences with a separable linear demand shifter; at interior thresholds, neutrality holds when that shifter is uniform. The corresponding weighted result is local; exact translation of the reduction-weighted type distribution remains specific to the quadratic benchmark. Additional comparative statics distinguish cap changes from changes in circumvention costs and access pricing.

Keywords: quantity caps; circumvention; regulatory targeting; heterogeneous demand; self-selection; online gaming

JEL classification: D11; L51; L86

1 Introduction

A quantity cap limits consumption only for users who remain subject to it. When unrestricted access can be restored at a private cost, tightening works through two margins: users who remain constrained consume less, while others may pay to restore unrestricted consumption. How does tightening a circumventable cap affect total consumption, and which users account for the resulting reductions?

Online gaming provides the motivating institutional setting. Wang (2026) finds that China’s 2021 restrictions reduced minors’ gaming engagement and Internet use in a high-school sample, with substantial but incomplete compliance. The Chinese rules restricted minors’ online gaming to designated windows on selected days and required real-name verification; the accompanying official explanation acknowledged circumvention using adult identities (National Press and Publication Administration 2021a,b). Feng et al. (2023) document account borrowing, purchasing, and time-priced rental. In a convenience sample of 430 adolescent heavy gamers, Zhou et al. (2024) find that greater reported pre-policy gaming time is associated with a higher probability of violating the restriction. Account acquisition or setup can contain a fixed component; the model isolates it over a common observation window.

The main result characterizes the aggregate effect of tightening a common cap with fixed-cost circumvention. In the quadratic benchmark, reductions arise over an intermediate interval of demand types. With both thresholds interior and the cost fixed, tightening shifts this interval downward: remaining constrained users consume less, while users crossing its upper boundary restore consumption discretely. A marginal tightening raises aggregate use to first order exactly when density at the circumvention threshold exceeds its average over the constrained interval. The reverse inequality implies a decrease, and equality gives a zero first-order response. Monotone-density comparisons also apply to finite cap changes within the interior region.

A second result concerns regulatory targeting at a fixed total reduction. In the quadratic benchmark with uniformly distributed demand types, cap changes at a fixed circumvention cost within the interior region leave the aggregate reduction unchanged. Tightening instead shifts the reduction-weighted type distribution toward lower-demand types. Consequently, the policy-weighted reduction weakly falls for every continuous, nonnegative, nondecreasing weight on demand. Higher-demand users who circumvent bear the private access cost but do not contribute to the aggregate consumption reduction.

The quadratic benchmark displays the moving interval directly; Proposition 3 establishes the scope of the aggregate result under a separable linear demand shifter. At interior thresholds, its envelope identity preserves the density condition and, when the linear demand shifter is uniformly distributed, exact aggregate neutrality and the local comparison for nondecreasing policy weights. The exact translation of the reduction-weighted type distribution in Proposition 2 is the stronger allocation property specific to the quadratic benchmark. Further comparisons return to quadratic utility: independent heterogeneous fixed costs preserve the uniform benchmark when every conditional threshold remains interior, while excess-use and alternative-account charges differ in which units of consumption are subject to the charge.

The closest quota analysis is [Haavio and Kotakorpi \(2016\)](#), who study sin licenses with preference uncertainty and secondary-market purchases. Their consumption rule allows quotas to bind at intermediate preference realizations while higher realizations lead to purchases beyond the quota. Their earlier working paper also considers a local maximum-quota intervention: with the secondary-market price unchanged, affected sophisticated buyers maintain total consumption while shifting purchases to the secondary market ([Haavio and Kotakorpi 2012](#), Sect. 5.3). The comparison here concerns population-wide consumption under a common cap, holding the private access cost fixed. In the uniform quadratic benchmark at interior thresholds, aggregate neutrality comes from consumption reductions among users who remain constrained offsetting the discrete restoration of consumption by switchers. The allocation result then characterizes which demand types account for the unchanged aggregate reduction.

The importance of alternative sources of access for observed policy responses also links the analysis to costly avoidance of tax and access restrictions. For cigarettes, [Lovenheim \(2008\)](#) studies cross-border purchasing, [Chiou and Muehlegger \(2008\)](#) model purchase-location choice together with consumption adjustment, and [DeCicca et al. \(2013\)](#) incorporate avoidance into the analysis of corrective taxation. [Lovenheim and Slemrod \(2010\)](#) examine evasion of minimum drinking-age restrictions through access in nearby jurisdictions.

The fixed-cost structure also connects to regulatory avoidance at size thresholds. [Sneeringer and Key \(2011\)](#) model the choice between reducing activity to a threshold and bearing a regulatory cost. [Ewens et al. \(2024\)](#) use the corresponding trade-off between distorted activity and regulatory costs to infer the cost of being public. This threshold-choice structure underlies the aggregate and allocation comparisons as the cap varies.

The targeting interpretation relates to costly participation and self-selection. [Nichols and Zeckhauser \(1982\)](#) and [Besley and Coate \(1992\)](#) analyze requirements that screen participation in transfer programs; [Alatas et al. \(2016\)](#) examine application costs and self-targeting experimentally. [Shepard and Wagner \(2025\)](#) distinguish sorting on private demand from targeting based on net social value in insurance markets. In the present setting, the cost is incurred to circumvent the restriction. Policy weights specify the criterion for comparing how consumption reductions are distributed across demand types.

Sections 2–4 develop the model, aggregate comparative statics, and allocation result. Section 5 examines separable demand shifters, cost heterogeneity, and access pricing; Section 6 concludes, and Appendices A–D provide boundary calculations and technical extensions.

2 Model and selection into circumvention

A unit mass of users has demand type $a \in [0, A]$, where $A > 0$. The type distribution F is continuously differentiable on $[0, A]$, satisfies $F(0) = 0$ and $F(A) = 1$, and has continuous density $f = F'$, positive on $(0, A)$. All population aggregates use this fixed distribution. The notation $\mathbb{E}_F$ denotes expectation with respect to F , and $\mathbf{1}\{\cdot\}$ is the

indicator function. Subscripts on functions denote partial derivatives with respect to the indicated arguments; a prime denotes an ordinary derivative.

A user chooses consumption $g \geq 0$, interpreted as gaming time over the observation window. In the baseline model, utility before any circumvention cost is

$$u(g; a) = ag - \frac{g^2}{2}. \quad (1)$$

Unrestricted consumption is $g^0(a) = a$. Type a measures private demand, including immediate opportunity costs. The common cap $c \geq 0$ restricts compliant consumption to $g \leq c$. It is a scalar restriction on total consumption over a common observation window; the timing of permitted access is outside this abstraction.

Paying a known, deterministic private cost $K > 0$, measured in utility units, fully restores unrestricted choice for that window. The cost is independent of subsequent use. Users know their types and costs and face no liquidity constraints. The parameters c and K are exogenous; comparisons vary them while holding the demand distribution and the remaining access technology fixed. Cap comparisons hold K fixed; Section 3.3 considers cost changes and joint parameter paths.

A user never benefits from paying K for consumption already feasible under compliance. The choice can therefore be written as $\max_{g \geq 0} \{u(g; a) - K \mathbf{1}\{g > c\}\}$. For $a > c$, the optimized compliant payoff is $ac - c^2/2$, while the optimized payoff after restoring access is $a^2/2 - K$. Their difference is

$$\left(\frac{a^2}{2} - K\right) - \left(ac - \frac{c^2}{2}\right) = \frac{(a - c)^2}{2} - K. \quad (2)$$

Write $s = \sqrt{2K}$ and $b = c + s$. Here b is the demand threshold for circumvention, and s is both the width of the constrained interval, when interior, and the consumption jump at its upper boundary.

Lemma 1 (Selection into circumvention) *In the quadratic model, let $c \geq 0$, $K > 0$, and $b = c + \sqrt{2K}$. Taking compliance at indifference, a user circumvents if and only if $a > b$. Consumption is*

$$g(a; c, K) = \begin{cases} a, & a \leq c, \\ c, & c < a \leq b, \\ a, & a > b, \end{cases} \quad a \in [0, A]. \quad (3)$$

Proof Strict concavity gives the unique unrestricted choice $g^0(a) = a$. If $a \leq c$, that choice is available without paying K . If $a > c$, the constrained optimum is c , and equation (2) is positive exactly when $a > c + \sqrt{2K}$. Comparing the two optimized payoffs gives equation (3). Atomlessness makes the tie convention irrelevant to aggregates. $\square$

The type-specific reduction is $r(a; c, K) = a - g(a; c, K) = (a - c) \mathbf{1}\{c < a \leq b\}$. Low-demand users choose their unrestricted quantities without circumventing. Intermediate-demand users face a binding cap. Higher-demand users pay K and restore

unrestricted consumption. Thus the types whose consumption is reduced form an interval rather than comprising all types above a demand cutoff.

For an interior configuration, $0 < c < b < A$, lowering c at fixed K shifts both endpoints of this interval downward by the same amount. Two distinct interior caps therefore generate intervals of equal width that are not nested. Remaining constrained users consume less, while users crossing the upper boundary restore unrestricted consumption. These two margins drive the aggregate response; the shifting interval also changes which types contribute to the aggregate reduction.

3 Cap stringency and aggregate consumption

For a fixed $K > 0$, the interior cap comparisons concern $0 < c < A - \sqrt{2K}$. In this region, unconstrained low-demand users, constrained users, and circumventers each have positive population mass. The neutrality comparisons below remain within this region; Appendix A.1 treats nonbinding caps, no-circumvention regimes, and their boundaries.

3.1 The local density condition

Let $Q(c, K) = \mathbb{E}_F[g(a; c, K)]$ denote aggregate consumption and $Q_0 = \mathbb{E}_F[a]$ its unrestricted benchmark. For $0 < c < b < A$, Lemma 1 gives

$$Q(c, K) = Q_0 - \int_c^b (a - c)f(a) da. \quad (4)$$

Differentiation separates the consumption change among constrained users from the consumption restored at the moving circumvention threshold.

Proposition 1 (Aggregate response to the cap) *In the quadratic model, let $s = \sqrt{2K}$ and $b = c + s$, with $K > 0$ and $0 < c < b < A$. Holding K and F fixed,*

$$Q_c = \underbrace{F(b) - F(c)}_{\text{constrained-user margin}} - \underbrace{sf(b)}_{\text{switching margin}}. \quad (5)$$

A marginal reduction in c increases aggregate consumption to first order if and only if

$$f(b) > \frac{F(b) - F(c)}{b - c}. \quad (6)$$

If f is nondecreasing on $[c, b]$, then $Q_c \leq 0$; if it is nonincreasing, then $Q_c \geq 0$. Under either monotonicity condition, the corresponding inequality is strict when f is nonconstant on that interval.

Proof Since $b_c = 1$, differentiation of equation (4) yields

$$Q_c = -(b - c)f(b)b_c + \int_c^b f(a) da = F(b) - F(c) - sf(b).$$

The lower-bound term vanishes because the reduction is zero at $a = c$. A marginal cap reduction has a positive first-order effect exactly when $Q_c < 0$. Using $s = b - c > 0$ gives

equation (6). A nondecreasing density has an endpoint value at least as large as its interval average; continuity and nonconstancy give a strict inequality. A nonincreasing density reverses this comparison. $\square$

For a small tightening $dc = -\varepsilon$, where $\varepsilon > 0$, each user who remains constrained reduces consumption by ε . At the same time, the upper threshold falls by ε , inducing approximately $f(b)\varepsilon$ additional users to circumvent. In the limit, each switch restores s units of consumption. Equation (5) compares the mass of constrained users contributing to the intensive-margin reduction with the mass entering circumvention multiplied by its consumption jump. An increasing density places relatively more users at the switching boundary, so lowering the cap can raise aggregate use.

3.2 Finite cap changes and neutrality under uniform types

The comparison also applies to finite changes when the relevant density condition holds along the entire path. Fix $K > 0$ and take two caps $0 < c_1 < c_2$ with $c_2 + s < A$. Substitution of $y = a - c$, where y is distance above the cap, gives

$$Q(c_1, K) - Q(c_2, K) = \int_0^s y[f(c_2 + y) - f(c_1 + y)] dy. \quad (7)$$

If f is nondecreasing throughout $[c_1, c_2 + s]$, lowering the cap from c_2 to c_1 weakly raises consumption; strict increase of f gives a strict increase in consumption. A nonincreasing density yields the reverse weak comparison, with strictness under strict decrease. These are finite comparisons within the interior region, rather than comparisons across support boundaries.

Exact neutrality under uniform types. With uniformly distributed types, $f(a) = 1/A$ on $[0, A]$, the two local margins offset. Integrating equation (4) gives

$$Q(c, K) = \frac{A}{2} - \frac{K}{A}, \quad 0 < c < c + s < A. \quad (8)$$

The entire reduction profile translates with c , retaining its width s , its shape, and its population density. Any cap change staying within this region leaves total consumption unchanged. Circumvention nevertheless increases as the cap falls: its population share is $E(c, K) = 1 - F(b)$, with $E_c = -f(b) < 0$. The same aggregate quantity can therefore coexist with different compliance patterns.

At every cap, baseline consumption satisfies $g(a; c, K) \leq a$, so $Q \leq Q_0$. An increase induced by tightening an existing cap is consistent with this bound. If $0 \leq c < A$ and $b > A$, no user circumvents and $Q_c = 1 - F(c) > 0$. If $c \geq A$, the cap is nonbinding for all users. Appendix A treats these regions, their boundaries, and examples of densities generating opposite interior signs.

3.3 Circumvention costs and joint parameter changes

The cap results hold the private cost fixed. A complementary comparison varies K , holding c and F fixed. At an interior configuration, $b_K = 1/s$, and differentiation of

equation (4) gives

$$Q_K = -(b - c)f(b)b_K = -f(b) < 0. \quad (9)$$

Increasing K moves the upper boundary of the constrained interval toward higher-demand users. Marginal circumventers switch to compliance, while the cap stays unchanged. The circumvention share satisfies $E_K = -f(b)/s < 0$. This monotonicity also holds globally: at fixed c , raising the cost can only move a user from unrestricted consumption to compliant consumption. Within the strict no-circumvention region, further increases in K leave consumption unchanged.

The two parameters can also change jointly. Let x vary over an interval and index a differentiable reform path $(c(x), K(x))$, with $c'(x) < 0$, $K'(x) \geq 0$, and $0 < c(x) < c(x) + \sqrt{2K(x)} < A$. Holding F fixed and evaluating s and b along this path, the chain rule gives

$$\frac{dQ}{dx} = [F(b) - F(c) - sf(b)]c'(x) - f(b)K'(x). \quad (10)$$

Since $-c'(x) > 0$ and $f(b) > 0$, the local change is positive exactly when

$$\frac{K'(x)}{-c'(x)} < s - \frac{F(b) - F(c)}{f(b)}. \quad (11)$$

A rise in the private cost counteracts the consumption increase from cap tightening when $Q_c < 0$. Under uniform types, equation (10) reduces to $dQ/dx = -K'(x)/A$. An enforcement interpretation of this parameter path requires specifying how the intervention changes the private cost of restoring access.

4 Regulatory targeting and the allocation of reductions

Aggregate quantity does not identify which users account for the aggregate reduction. In the quadratic benchmark with uniform types, cap changes at fixed K within the interior region leave the total reduction unchanged. Policy weights and the reduction-weighted distribution of demand types describe how its allocation changes.

Let $w : [0, A] \rightarrow [0, \infty)$ be a continuous policy weight, fixed independently of c and K . It assigns importance to reducing one unit of consumption by a user of type a . Define weighted consumption and its unrestricted benchmark by

$$H(c, K) = \int_0^A w(a)g(a; c, K)f(a) da, \quad H_0 = \int_0^A w(a)af(a) da. \quad (12)$$

Then $H_0 - H$ is the policy-weighted reduction. Nondecreasing w describes a policy that gives weakly greater priority to reducing consumption by higher-demand users. The weights specify an allocation criterion; demand intensity is not equated with addiction or social harm. A welfare comparison would additionally require the benefits of consumption and the relevant private and social costs. In each specification, Q_0 and H_0 denote the stated unrestricted reference levels.

For an arbitrary density and an interior threshold, weighted reductions satisfy $H_0 - H = \int_c^b w(a)(a - c)f(a) da$. Differentiating as in Proposition 1, with the other policy parameter, F , and w held fixed in each partial derivative, gives

$$H_c = \int_c^b w(a)f(a) da - sw(b)f(b), \quad H_K = -w(b)f(b) \leq 0. \quad (13)$$

Thus the local weighted comparison depends on the product wf . Under uniform types, the allocation can be described directly by a reduction-weighted distribution of demand types. Fix $K > 0$ and an interior cap $0 < c < c + s < A$. Normalizing the type-specific reductions by their aggregate amount gives the density

$$p_c(a) = \frac{r(a; c, K)f(a)}{Q_0 - Q(c, K)} = \frac{a - c}{K} \mathbf{1}\{c < a < c + s\}. \quad (14)$$

Endpoint choices do not affect this density. It integrates to one because $s^2/2 = K$. Let Z be a random variable with density z/K for $0 < z < s$, and zero otherwise. The density p_c is that of $c + Z$. Lowering the cap therefore translates this reduction-weighted type distribution toward lower-demand types, while leaving its total mass unchanged. This distribution weights users by their consumption reductions rather than counting each user equally.

Proposition 2 (Reallocation of consumption reductions) *In the quadratic model with uniformly distributed types, fix $K > 0$, write $s = \sqrt{2K}$, and take two caps $0 < c_1 < c_2$ with $c_2 + s < A$. Both caps generate the aggregate reduction K/A . For every continuous, nonnegative, nondecreasing weight w , the policy-weighted reduction at the tighter cap c_1 is no greater than at c_2 ; equivalently,*

$$H(c_1, K) \geq H(c_2, K). \quad (15)$$

The inequality is strict if w is strictly increasing on $[c_1, c_2 + s]$.

Proof Equation (8) gives the common aggregate reduction. For either cap, substitution of $z = a - c$, where z is distance above that cap, yields

$$H_0 - H(c, K) = \frac{1}{A} \int_0^s w(c + z)z dz. \quad (16)$$

For every $z \in [0, s]$, monotonicity gives $w(c_1 + z) \leq w(c_2 + z)$. Integrating against the nonnegative weight z/A proves equation (15). Under strict increase of w , the integrand inequality is strict for every $z \in (0, s)$, a set of positive measure. $\square$

For illustration, take the parameter values $A = 2$, $K = 1/8$, and $w(a) = a$. Reducing the cap from 1 to $1/2$ increases the circumvention share E , preserves total consumption, and raises weighted consumption (Table 1).

For this weight, equation (16) gives $H = A^2/3 - [cK + (2K)^{3/2}/3]/A$, with $H_c = -K/A$.

Under the conditions of Proposition 2, tightening reallocates an unchanged total reduction away from types receiving higher policy weights when policy priority rises

Table 1 Exact illustration with $A = 2$, $K = 1/8$, and $w(a) = a$

Quantity	Cap $c = 1$	Cap $c = 1/2$
Circumvention threshold b	$3/2$	1
Constrained interval	$(1, 3/2]$	$(1/2, 1]$
Circumvention share E	$1/4$	$1/2$
Aggregate consumption Q	$15/16$	$15/16$
Weighted consumption H	$5/4$	$41/32$

with demand. Increasing K , by contrast, extends the constrained interval toward higher demand and weakly reduces both Q and H .

5 Scope of the aggregate characterization

Section 5.1 allows nonquadratic curvature while retaining the linear demand shifter and fixed-cost access. It establishes the scope of the aggregate density condition and neutrality, together with the local weighted comparison. Section 5.2 returns to quadratic utility and varies cost heterogeneity and the pricing of restored access.

5.1 Separable linear demand shifters

Retain the type distribution and fixed-cost circumvention, and replace equation (1) by

$$u(g; a) = ag - v(g), \quad (17)$$

where $v : [0, \infty) \rightarrow \mathbb{R}$ is twice continuously differentiable, $v(0) = v'(0) = 0$, $v''(g) > 0$, and $v'(g) \rightarrow \infty$ as $g \rightarrow \infty$. Unrestricted consumption and the type whose unrestricted choice equals the cap are

$$d(a) = (v')^{-1}(a), \quad \alpha = v'(c), \quad (18)$$

where $0 < c < d(A)$. Hence $d(\alpha) = c$. The uniform benchmark in this subsection imposes uniformity on the linear demand shifter a , rather than on unrestricted consumption $d(a)$. It is a joint restriction on the utility parameterization and the type distribution. When d is nonlinear, unrestricted consumption is not uniformly distributed. In this section, unrestricted benchmarks are $Q_0 = \mathbb{E}_F[d(a)]$ and $H_0 = \mathbb{E}_F[w(a)d(a)]$; the definitions of Q and H as actual aggregate and weighted consumption are unchanged.

For $a \geq \alpha$, let the gain from unrestricted access before paying K be

$$\Delta(a, c) = ad(a) - v(d(a)) - ac + v(c). \quad (19)$$

The envelope identities are

$$\Delta_a(a, c) = d(a) - c, \quad \Delta_c(a, c) = -(a - \alpha). \quad (20)$$

Since $\Delta(\alpha, c) = 0$ and $\Delta_a > 0$ for $a > \alpha$, the restriction $0 < K < \Delta(A, c)$ yields a unique circumvention threshold $b \in (\alpha, A)$ satisfying $\Delta(b, c) = K$. Types below α consume $d(a)$; types in $(\alpha, b]$ consume c ; and types above b consume $d(a)$. Here b is defined by this indifference condition, rather than by the baseline formula $c + s$. Implicit differentiation gives

$$b_c = \frac{b - \alpha}{d(b) - c}, \quad b_K = \frac{1}{d(b) - c}. \quad (21)$$

Both derivatives are positive. The aggregate reduction is

$$Q_0 - Q(c, K) = \int_{\alpha}^b [d(a) - c] f(a) da. \quad (22)$$

Proposition 3 (Density condition and neutrality with a separable linear demand shifter)
Consider $u(g; a) = ag - v(g)$, where $v : [0, \infty) \rightarrow \mathbb{R}$ is twice continuously differentiable, $v(0) = v'(0) = 0$, $v''(g) > 0$, and $v'(g) \rightarrow \infty$ as $g \rightarrow \infty$. Let F be a continuously differentiable distribution on $[0, A]$, with $A > 0$, $F(0) = 0$, $F(A) = 1$, and continuous density $f = F'$, positive on $(0, A)$.

Under fixed-cost circumvention, write $d(a) = (v')^{-1}(a)$, $\alpha = v'(c)$, and $\Delta(a, c) = ad(a) - v(d(a)) - ac + v(c)$. Take $0 < c < d(A)$ and $0 < K < \Delta(A, c)$, and let $b \in (\alpha, A)$ solve $\Delta(b, c) = K$. In each partial derivative, the other policy parameter, F , and v are held fixed. Then

$$Q_c = F(b) - F(\alpha) - (b - \alpha)f(b), \quad Q_K = -f(b). \quad (23)$$

A marginal cap reduction increases aggregate consumption to first order exactly when $f(b) > [F(b) - F(\alpha)]/(b - \alpha)$. For any continuous, nonnegative weight w fixed independently of c and K ,

$$H_c = \int_{\alpha}^b w(a)f(a) da - (b - \alpha)w(b)f(b), \quad H_K = -w(b)f(b). \quad (24)$$

If the linear demand shifter a is uniform on $[0, A]$,

$$Q_0 - Q(c, K) = \frac{K}{A}. \quad (25)$$

In that case, $H_c \leq 0$ for nondecreasing w , with strict inequality if w is strictly increasing on $[\alpha, b]$.

Proof Strict convexity of v gives the unique choice $d(a)$. Differentiating equation (19), using $v'(d(a)) = a$, establishes equation (20). Differentiating $\Delta(b, c) = K$ gives equation (21). Leibniz's rule applied to equation (22) yields

$$Q_c = \int_{\alpha}^b f(a) da - [d(b) - c]f(b)b_c.$$

The lower-bound term is zero because $d(\alpha) = c$. Substituting b_c proves the first part of equation (23). Differentiation with respect to K and substitution of b_K prove its second part. Applying the same argument to the weighted reduction proves equation (24).

For uniform types, integrate $\Delta_a = d(a) - c$ over the constrained interval:

$$Q_0 - Q = \frac{1}{A} \int_{\alpha}^b [d(a) - c] da = \frac{\Delta(b, c) - \Delta(\alpha, c)}{A} = \frac{K}{A}.$$

Finally, under uniformity, equation (24) becomes $H_c = A^{-1}\{\int_{\alpha}^b w(a) da - (b - \alpha)w(b)\}$. Monotonicity of w gives the asserted sign, with strictness under strict increase. $\square$

The width of the constrained type interval, $b - \alpha$, and the consumption jump, $d(b) - c$, need not coincide. Equation (21) links the speed of the moving boundary to their ratio; multiplying that speed by the consumption jump recovers $b - \alpha$. This is why the same density comparison survives.

When the demand shifter a is uniformly distributed and policy weights are non-decreasing, a marginal tightening weakly lowers the policy-weighted reduction to first order at an interior configuration. This is the local weighted comparison; the exact translation in Proposition 2 remains specific to the quadratic model. Appendix B gives the corresponding single-crossing decomposition under interior unrestricted choices.

5.2 Cost heterogeneity and alternative access pricing

Cost heterogeneity preserves the uniform benchmark under a specific independence condition. Return to quadratic utility and let the private cost k vary independently of a , with distribution G on $[\underline{k}, \bar{k}] \subset (0, \infty)$. If $c + \sqrt{2\bar{k}} < A$ and $c > 0$, every conditional threshold is interior. Under uniform types, integrating equation (8) over costs gives $Q = A/2 - \mathbb{E}_G[k]/A$, where $\mathbb{E}_G$ denotes expectation over G . For nondecreasing policy weights, a marginal tightening weakly lowers the policy-weighted reduction to first order, holding the cost distribution fixed. Appendix C provides the derivatives and explains why correlation between demand and costs, or binding support boundaries, changes the scope of the result.

The set of consumption units subject to the charge matters separately from whether the charge is fixed or per unit. Let $\tau \geq 0$ be a price in utility units per unit of consumption. First, users may retain their compliant allowance and pay only for consumption above the cap; write Q_{ex} for aggregate consumption in this arrangement. With no fixed entry cost, consumption joins continuously at the upper type threshold, eliminating the discrete switching term. A positive fixed cost retains the jump. With uniformly distributed demand types and an interior threshold, $\partial Q_{\text{ex}}/\partial c = \tau/A$, holding K , τ , and F fixed. A positive per-unit price therefore breaks uniform neutrality.

Alternatively, circumvention may replace compliant access, with the per-unit charge applying to all consumption on the alternative account; write Q_{alt} for its aggregate consumption. Switching can then produce a consumption jump even without a fixed entry fee. Under uniform types at an interior threshold, the cap derivative is zero, as equation (D.10) shows. The comparison turns on the consumption jump multiplied by the speed of the moving threshold: this switching term exactly offsets the constrained-user margin when the per-unit charge applies to all consumption on the alternative account. Appendix D derives both cases.

6 Discussion and conclusion

At a fixed circumvention cost, a tighter cap changes both use among constrained users and which demand types remain constrained. Within the interior region, the aggregate response depends on density at the circumvention boundary relative to its average

over the constrained interval. Uniform types yield exact neutrality; in the quadratic model, tightening then reallocates the same total reduction toward lower-demand types. The policy-weighted reduction weakly falls for every continuous, nonnegative, nondecreasing weight.

The density condition extends to the separable linear-shifter class; under a uniform shifter, exact neutrality also holds within the interior region. The exact translation of the reduction-weighted type distribution is the sharper allocation feature of the quadratic benchmark. Increasing the private cost of circumvention instead extends the constrained interval toward higher demand and weakly lowers consumption. Tightening the cap and raising the cost of circumvention affect selection into the constrained group through different margins.

Online gaming provides the motivating application for fixed-cost circumvention and demand-based selection. Empirical assessment requires information on actual consumption, circumvention, the underlying demand distribution, and the costs and conditions of alternative access, including heterogeneity in private costs. Observed post-policy consumption bunches at the cap, so the resulting consumption distribution cannot be interpreted directly as the underlying demand distribution. Evaluating the restriction therefore requires distinguishing the total consumption reduction from its allocation across demand types.

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Appendix A Boundary cases and additional distributional calculations

A.1 Support boundaries and cost limits

This appendix uses the quadratic model. If $c \geq A$, all users can choose unrestricted consumption without circumventing, so $Q = Q_0$, $H = H_0$, and $E = 0$. If $0 \leq c < A$

and $c + s \geq A$, no positive mass circumvents. Consumption is $\min\{a, c\}$, and

$$\begin{aligned} Q &= \int_0^c af(a) da + c[1 - F(c)], \\ H &= \int_0^c w(a)af(a) da + c \int_c^A w(a)f(a) da. \end{aligned} \quad (\text{A.1})$$

In the strict region $c + s > A$, with $0 < c < A$,

$$Q_c = 1 - F(c) > 0, \quad H_c = \int_c^A w(a)f(a) da \geq 0, \quad Q_K = H_K = 0. \quad (\text{A.2})$$

At $c + s = A$, the consumption levels are continuous, but one-sided derivatives from the two regimes may differ when $f(A) > 0$. At $c = 0$, cap derivatives are right derivatives. Interior formulas should therefore be used only within their stated regime.

When $K = 0$, every type above the cap can restore unrestricted access at no cost, giving $g(a) = a$, $Q = Q_0$, and $H = H_0$. As $K \downarrow 0$, the constrained interval shrinks and the same limits follow by dominated convergence: $0 \leq g(a; c, K) \leq A$, and w is bounded on the compact type support. For fixed c , sufficiently large K produces the no-circumvention regime.

At every cap and cost, $0 \leq g(a; c, K) \leq a$. Also, raising K at fixed c can only move a type from unrestricted use to compliant use. These observations establish $Q \leq Q_0$, $H \leq H_0$, and global weak monotonicity of both aggregates in K , including across support boundaries.

A.2 Densities with opposite signs and the low-cost limit

For the increasing density $f(a) = 2a/A^2$ on $[0, A]$, direct integration of equation (4) yields

$$Q = \frac{2A}{3} - \frac{cs^2 + 2s^3/3}{A^2}, \quad Q_c = -\frac{s^2}{A^2} < 0. \quad (\text{A.3})$$

For the decreasing density $f(a) = 2(A - a)/A^2$,

$$Q = \frac{A}{3} - \frac{(A - c)s^2 - 2s^3/3}{A^2}, \quad Q_c = \frac{s^2}{A^2} > 0. \quad (\text{A.4})$$

Both expressions require $0 < c < c + s < A$. They use the same preferences and access technology, with the sign changing only through the demand distribution.

For a fixed $c \in (0, A)$, suppose f is twice continuously differentiable in a neighborhood of c . Taylor expansion in $s = \sqrt{2K}$ gives

$$Q_0 - Q = Kf(c) + O(K^{3/2}), \quad Q_c = -Kf'(c) + O(K^{3/2}) \quad \text{as } K \downarrow 0. \quad (\text{A.5})$$

Here $O(K^{3/2})$ denotes a remainder bounded in absolute value by a constant times $K^{3/2}$ for sufficiently small positive K , with c fixed. The expansion follows by integrating $yf(c + y)$ and by subtracting $sf(c + s)$ from $\int_0^s f(c + y) dy$. If

$m(a) = w(a)f(a)$ is twice continuously differentiable near c , the same argument gives $H_c = -Km'(c) + O(K^{3/2})$. Cheap circumvention makes the total reduction small while leaving its cap derivative sensitive to local demand density.

Appendix B Single-crossing comparisons with interior unrestricted choices

Retain the type distribution, positive cost K , and weights defined in the main text. Let $u(g; a)$ be twice continuously differentiable for $g \geq 0$ and $a \in [0, A]$, strictly concave in g , and satisfy $u_{ga}(g; a) > 0$ on this domain. Assume a unique, continuous, interior unrestricted choice $d(a) > 0$ for each $a \in [0, A]$. Suppose an interior type $\alpha \in (0, A)$ satisfies $d(\alpha) = c$, and a threshold $b \in (\alpha, A)$ satisfies $\Delta(b, c) = K$, where $\Delta(a, c) = u(d(a); a) - u(c; a)$. These are the regularity and interiority conditions used in this appendix.

Cap derivatives hold K , F , u , and w fixed; cost derivatives hold c , F , u , and w fixed.

The unrestricted first-order condition is $u_g(d(a); a) = 0$. At a lower type's optimum, a higher type has strictly positive marginal utility because $u_{ga} > 0$. Strict concavity therefore implies that d is strictly increasing. The envelope theorem gives, for $a > \alpha$,

$$\begin{aligned}\Delta_a(a, c) &= u_a(d(a); a) - u_a(c; a) = \int_c^{d(a)} u_{ga}(t; a) dt > 0, \\ \Delta_c(a, c) &= -u_g(c; a) < 0,\end{aligned}\tag{B.1}$$

where t is a consumption integration variable. The gain is strictly increasing above α , so b is unique. The equation $u_g(c; \alpha) = 0$ defines a differentiable lower threshold because its derivative with respect to α is $u_{ga} > 0$. Likewise, $\Delta_a(b, c) > 0$ permits implicit differentiation of $\Delta(b, c) = K$, giving

$$b_c = \frac{u_g(c; b)}{\Delta_a(b, c)} > 0, \quad b_K = \frac{1}{\Delta_a(b, c)} > 0.\tag{B.2}$$

Define $Q_0 = \mathbb{E}_F[d(a)]$ and $H_0 = \mathbb{E}_F[w(a)d(a)]$. The compliant choice is $\min\{d(a), c\}$; hence only types above α compare $\Delta(a, c)$ with K . Types in $(\alpha, b]$ remain constrained, and types above b restore unrestricted consumption. The reduction integral is $Q_0 - Q = \int_\alpha^b [d(a) - c]f(a) da$. Differentiating it, with the lower-bound term zero, yields

$$Q_c = F(b) - F(\alpha) - [d(b) - c]f(b)b_c, \quad Q_K = -\frac{[d(b) - c]f(b)}{\Delta_a(b, c)} < 0.\tag{B.3}$$

The corresponding weighted derivatives are

$$\begin{aligned} H_c &= \int_{\alpha}^b w(a)f(a) da - [d(b) - c]w(b)f(b)b_c, \\ H_K &= -\frac{[d(b) - c]w(b)f(b)}{\Delta_a(b, c)} \leq 0. \end{aligned} \tag{B.4}$$

The intensive and switching margins remain. Outside the separable-shifter class, the product $[d(b) - c]b_c$ need not equal $b - \alpha$, so the interval-average density condition and uniform neutrality do not generally follow.

Appendix C Independent heterogeneous fixed costs

Return to quadratic utility. A user's private cost k has distribution G on the compact interval $[\underline{k}, \bar{k}] \subset (0, \infty)$, independently of a . The distribution G may have atoms. Assume $0 < c < c + \sqrt{2\bar{k}} < A$. For each cost, write $s_k = \sqrt{2k}$ and $b_k = c + s_k$. Conditional choices are those of Lemma 1 with $K = k$.

Let $Q_G(c)$ and $H_G(c)$ denote consumption aggregated over both demand and costs. In terms of the baseline functions,

$$Q_G(c) = \int Q(c, k) dG(k), \quad H_G(c) = \int H(c, k) dG(k). \tag{C.1}$$

All integrals with respect to G are over its support. Holding F , G , and w fixed, the cap derivatives are

$$\begin{aligned} Q'_G(c) &= \int [F(b_k) - F(c) - s_k f(b_k)] dG(k), \\ H'_G(c) &= \int \left[\int_c^{b_k} w(a)f(a) da - s_k w(b_k)f(b_k) \right] dG(k). \end{aligned} \tag{C.2}$$

Continuity of f and w , compact cost support bounded away from zero, and the strict interior condition bound the conditional derivatives uniformly in a neighborhood of c ; hence differentiation under the integral is valid.

With uniformly distributed demand types,

$$Q_G(c) = \frac{A}{2} - \frac{\mathbb{E}_G[k]}{A}, \quad Q'_G(c) = 0. \tag{C.3}$$

Nondecreasing w gives $H'_G(c) \leq 0$. Strict increase of w over the union of constrained intervals makes this inequality strict. For $w(a) = a$,

$$H_G(c) = \frac{A^2}{3} - \frac{c\mathbb{E}_G[k] + \mathbb{E}_G[(2k)^{3/2}]/3}{A}, \quad H'_G(c) = -\frac{\mathbb{E}_G[k]}{A}. \tag{C.4}$$

For a common additive cost increase $\lambda \geq 0$, define $Q_G(c, \lambda) = \int Q(c, k + \lambda) dG(k)$ and $H_G(c, \lambda) = \int H(c, k + \lambda) dG(k)$, so the one-argument functions correspond to $\lambda = 0$. Holding c , F , G , and w fixed, the cost-shift derivative satisfies, whenever $c + \sqrt{2(\bar{k} + \lambda)} < A$,

$$\frac{\partial Q_G(c, \lambda)}{\partial \lambda} = - \int f\left(c + \sqrt{2(k + \lambda)}\right) dG(k) \leq 0. \quad (\text{C.5})$$

The derivative for H_G replaces f by wf . At $\lambda = 0$, these derivatives can be interpreted as right derivatives. Global weak monotonicity follows type by type even when some conditional thresholds cross the support boundary.

Each cost group has its own threshold b_k , whether or not demand and cost are independent. Independence makes the conditional demand distribution equal to F in every group, permitting the baseline formulas to be averaged over G . With correlation, those conditional distributions need not coincide. If some conditional thresholds reach A , the corresponding formulas in Appendix A apply. Equation (C.3) is therefore an interior independence benchmark.

Appendix D Alternative charges for circumvented access

This appendix retains quadratic utility and the demand distribution from Section 2. The price $\tau \geq 0$ is measured in utility units per unit of consumption. The labels *ex* and *alt* distinguish charges on excess use from charges on all use on an alternative account. In both cases, the access arrangement and prices are held fixed when the cap changes. All partial derivatives below hold the other two policy parameters and F fixed; weighted derivatives also hold w fixed. Define $Q_j(c, K, \tau) = \mathbb{E}_F[g_j(a; c, K, \tau)]$ and $H_j(c, K, \tau) = \mathbb{E}_F[w(a)g_j(a; c, K, \tau)]$ for the corresponding consumption rule g_j , where $j \in \{\text{ex}, \text{alt}\}$. Arguments are suppressed below when their meaning is clear. Their unrestricted reference levels remain $Q_0 = \mathbb{E}_F[a]$ and $H_0 = \mathbb{E}_F[w(a)a]$.

D.1 Charging only for consumption above the cap

Suppose the user retains the compliant allowance. After paying $K > 0$, consumption $g \geq c$ yields payoff $u(g; a) - K - \tau(g - c)$. For $a > c + \tau$, the conditional optimum is $g = a - \tau$, and its gain over compliance is

$$\frac{(a - c - \tau)^2}{2} - K. \quad (\text{D.1})$$

Types with $a \leq c + \tau$ have no profitable above-cap choice. Thus the circumvention threshold is $b_{\text{ex}} = c + \tau + s$, where $s = \sqrt{2K}$. The rule g_{ex} equals a below c , c on $(c, b_{\text{ex}}]$, and $a - \tau$ above b_{ex} . Consumption jumps by s at the upper threshold.

For $0 < c < b_{\text{ex}} < A$,

$$Q_{\text{ex}} = Q_0 - \int_c^{b_{\text{ex}}} (a - c)f(a) da - \tau[1 - F(b_{\text{ex}})]. \quad (\text{D.2})$$

Differentiation gives

$$\begin{aligned} \frac{\partial Q_{\text{ex}}}{\partial c} &= F(b_{\text{ex}}) - F(c) - sf(b_{\text{ex}}), \\ \frac{\partial Q_{\text{ex}}}{\partial K} &= -f(b_{\text{ex}}), \\ \frac{\partial Q_{\text{ex}}}{\partial \tau} &= -[1 - F(b_{\text{ex}})] - sf(b_{\text{ex}}). \end{aligned} \quad (\text{D.3})$$

The weighted cap derivative is

$$\frac{\partial H_{\text{ex}}}{\partial c} = \int_c^{b_{\text{ex}}} w(a)f(a) da - sw(b_{\text{ex}})f(b_{\text{ex}}). \quad (\text{D.4})$$

The constrained interval has width $\tau + s$, while the consumption jump remains s . With uniform types, $\partial Q_{\text{ex}}/\partial c = \tau/A$. When $K = 0$, consumption becomes continuous at $b_{\text{ex}} = c + \tau$, and, for $0 < c < c + \tau < A$,

$$\frac{\partial Q_{\text{ex}}}{\partial c} = F(c + \tau) - F(c) \geq 0, \quad \frac{\partial H_{\text{ex}}}{\partial c} = \int_c^{c+\tau} w(a)f(a) da \geq 0. \quad (\text{D.5})$$

If $K = \tau = 0$, consumption is unrestricted and both cap derivatives are zero. A purely marginal charge on excess consumption therefore removes the switching jump. A positive K retains the possibility of a reversal even when $\tau > 0$: for $f(a) = 2a/A^2$, equation (D.3) gives

$$\frac{\partial Q_{\text{ex}}}{\partial c} = \frac{2c\tau + \tau^2 - s^2}{A^2}, \quad (\text{D.6})$$

which is negative exactly when $2K > 2c\tau + \tau^2$, subject to the interior condition.

D.2 Charging for all consumption on an alternative account

Now circumvention replaces compliant access. The user pays $K + \tau g$ on all consumption through an alternative account. Allow $K \geq 0$, $\tau \geq 0$, and take $c > 0$ with $K + \tau c > 0$. For a user who chooses this account, $a > c + \tau$, the conditional optimum is $g = a - \tau$, and its gain over compliance is

$$\frac{(a - c - \tau)^2}{2} - \tau c - K. \quad (\text{D.7})$$

For types with $a \leq c + \tau$, any conditional optimum at or below c is already feasible without the additional charges, so these types do not choose circumvention. Define

$$s_{\text{alt}} = \sqrt{2(K + \tau c)}, \quad b_{\text{alt}} = c + \tau + s_{\text{alt}}. \quad (\text{D.8})$$

Taking compliance at indifference, g_{alt} is a below c , c on $(c, b_{\text{alt}}]$, and $a - \tau$ above b_{alt} . The jump is $s_{\text{alt}} > 0$. Also, $\partial b_{\text{alt}}/\partial c = 1 + \tau/s_{\text{alt}}$ and $\partial b_{\text{alt}}/\partial K = 1/s_{\text{alt}}$. For $0 < c < b_{\text{alt}} < A$, the aggregate level is equation (D.2) with b_{ex} replaced by b_{alt} . Consequently,

$$\begin{aligned}\frac{\partial Q_{\text{alt}}}{\partial c} &= F(b_{\text{alt}}) - F(c) - (b_{\text{alt}} - c)f(b_{\text{alt}}), \\ \frac{\partial H_{\text{alt}}}{\partial c} &= \int_c^{b_{\text{alt}}} w(a)f(a) da - (b_{\text{alt}} - c)w(b_{\text{alt}})f(b_{\text{alt}}), \\ \frac{\partial Q_{\text{alt}}}{\partial K} &= -f(b_{\text{alt}}).\end{aligned}\tag{D.9}$$

The switching term contains $s_{\text{alt}}(1 + \tau/s_{\text{alt}}) = b_{\text{alt}} - c$. Under uniform types, direct integration gives

$$Q_{\text{alt}} = \frac{A}{2} - \tau - \frac{K - \tau^2/2}{A}, \quad 0 < c < c + \tau + \sqrt{2(K + \tau c)} < A. \tag{D.10}$$

Uniform neutrality therefore holds despite a per-unit charge. If $K = 0$, $\tau > 0$, and $c > 0$, the jump persists because switching entails paying for units available without that charge under compliance. Equations (D.5) and (D.9) show why a per-unit price must be interpreted together with the set of units it prices.

References

- Alatas V, Banerjee A, Hanna R, et al (2016) Self-targeting: Evidence from a field experiment in Indonesia. *Journal of Political Economy* 124:371–427. <https://doi.org/10.1086/685299>
- Besley T, Coate S (1992) Workfare versus welfare: Incentive arguments for work requirements in poverty-alleviation programs. *American Economic Review* 82:249–261. URL <https://www.jstor.org/stable/2117613>
- Chiou L, Muehlegger E (2008) Crossing the line: Direct estimation of cross-border cigarette sales and the effect on tax revenue. *The BE Journal of Economic Analysis & Policy* 8:Article 48. <https://doi.org/10.2202/1935-1682.2027>
- DeCicca P, Kenkel D, Liu F (2013) Excise tax avoidance: The case of state cigarette taxes. *Journal of Health Economics* 32:1130–1141. <https://doi.org/10.1016/j.jhealeco.2013.08.005>
- Ewens M, Xiao K, Xu T (2024) Regulatory costs of being public: Evidence from bunching estimation. *Journal of Financial Economics* 153:103775. <https://doi.org/10.1016/j.jfineco.2023.103775>

- Feng Y, Zhai R, Sion R, et al (2023) A study of China’s censorship and its evasion through the lens of online gaming. In: 32nd USENIX Security Symposium (USENIX Security 23). USENIX Association, pp 2599–2616, URL <https://www.usenix.org/conference/usenixsecurity23/presentation/feng>
- Haavio M, Kotakorpi K (2012) Sin licenses revisited. Discussion Paper No. 75, Aboa Centre for Economics, URL <https://www.ace-economics.fi/kuvat/dp75.pdf>
- Haavio M, Kotakorpi K (2016) Sin licenses revisited. *Journal of Public Economics* 144:40–51. <https://doi.org/10.1016/j.jpubeco.2016.09.006>
- Lovenheim MF (2008) How far to the border?: The extent and impact of cross-border casual cigarette smuggling. *National Tax Journal* 61:7–33. <https://doi.org/10.17310/ntj.2008.1.01>
- Lovenheim MF, Slemrod J (2010) The fatal toll of driving to drink: The effect of minimum legal drinking age evasion on traffic fatalities. *Journal of Health Economics* 29:62–77. <https://doi.org/10.1016/j.jhealeco.2009.10.001>
- National Press and Publication Administration (2021a) Notice on further strict management to prevent minors’ addiction to online games. Notice No. 14, 30 August, URL https://www.nppa.gov.cn/xxfb/tzgs/202108/t20210830_666285.html, In Chinese; title translated
- National Press and Publication Administration (2021b) Official explanation of the notice on preventing minors’ addiction to online games. 30 August, URL https://www.nppa.gov.cn/xxgk/zcjd/202108/t20210830_4759.html, In Chinese; title translated
- Nichols AL, Zeckhauser RJ (1982) Targeting transfers through restrictions on recipients. *American Economic Review* 72:372–377. URL <https://www.jstor.org/stable/1802361>
- Shepard M, Wagner M (2025) Do ordeals work for selection markets? Evidence from health insurance auto-enrollment. *American Economic Review* 115:772–822. <https://doi.org/10.1257/aer.20231133>
- Sneeringer S, Key N (2011) Effects of size-based environmental regulations: Evidence of regulatory avoidance. *American Journal of Agricultural Economics* 93:1189–1211. <https://doi.org/10.1093/ajae/aar040>
- Wang Z (2026) Restricting video games in China: Effects on time use, educational achievement, and health. *Journal of Development Economics* 182:103812. <https://doi.org/10.1016/j.jdeveco.2026.103812>

Zhou X, Liao M, Gorowska M, et al (2024) Compliance and alternative behaviors of heavy gamers in adolescents to Chinese online gaming restriction policy. *Journal of Behavioral Addictions* 13:687–692. <https://doi.org/10.1556/2006.2024.00021>